Evaluation of Differentials and Heights of Modular Jacobians

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Abstract

We bound the stable Faltings height of a Jacobian by the discriminant of one unramified fibre of a Belyi map. The proof evaluates integral regular differentials in a trace-dual lattice and estimates the complex evaluation maps by the mean-value inequality. For a modular curve of degree d over the j-line and level N, the different of the elliptic torsion scheme gives an explicit bound of order $d\log(2N)$, uniformly in the subgroup defining the curve. Applied to the modular realization underlying the five-contact Diophantine construction, it yields

$$h(a) < 2^{330770} M_5(a)^3 (1 + \log M_5(a)) < 2^{330789} M_5(a)^{7/2}.$$

The height h is logarithmic. The geometric and modular realization is stated as the arithmetic input; no additional *abc* or Szpiro hypothesis is imposed. All bounds use the original stable Faltings metric, and the numerical constants admit exact verification.

1 Conventions and the evaluation estimate

If K is a number field, $\mathcal{O}_K$ denotes its ring of integers. For a nonzero ideal $\mathfrak{a} \subset \mathcal{O}_K$, write $\mathrm{N}\mathfrak{a} = \#(\mathcal{O}_K/\mathfrak{a})$. If E is a projective $\mathcal{O}_K$-module with Hermitian metrics at its complex embeddings, $\widehat{\deg}(E)$ means the arithmetic degree of its determinant. In rank one, for a nonzero integral section s,

$$\widehat{\deg}(E) = \log \#(E/\mathcal{O}_K s) - \sum_{\sigma : K \hookrightarrow \mathbb{C}} \log \|s\|_\sigma.$$

The definition is independent of the choice of s by the product formula. Both embeddings in a complex conjugate pair are included.

Throughout, h_{F} is the original stable Faltings height. For a curve X/K with semistable reduction, its Jacobian height is

$$h_{\mathrm{F}}(\mathrm{Jac}\, X) = \frac{\widehat{\deg} H^0(\mathcal{X}, \omega_{\mathcal{X}/\mathcal{O}_K})}{[K : \mathbb{Q}]}, \tag{1}$$

where $\mathcal{X}$ is a minimal regular semistable model. The metric on differentials at a complex embedding is

$$\langle \omega, \eta \rangle = \frac{i}{2} \int_{X(\mathbb{C})} \omega \wedge \overline{\eta}.$$

This is the determinant metric corresponding to the original Faltings normalization; see [2, Sections 2.1 and 2.3] and [4, Section 2.1]. Passing to a finite extension on which the curve is semistable does not change the stable height.

A *Belyi map* is a finite morphism from a smooth projective geometrically connected curve to $\mathbb{P}^1$, unramified outside $0, 1, \infty$. Let

$$\pi : X \longrightarrow \mathbb{P}^1$$

be such a map, of degree d, where X has genus $g \geq 1$. Write t for the coordinate on the target. Choose a number field K over which X and π are defined, X is semistable, and every point of $\pi^{-1}(2)$ is rational. Let $\mathcal{Y}$ be the normalization of $\mathbb{P}^1_{\mathcal{O}_K}$ in $K(X)$. It is finite and flat over $\mathbb{P}^1_{\mathcal{O}_K}$. On the affine t-chart write its algebra as B, a finite projective $\mathcal{O}_K[t]$-algebra of rank d, and put

$$C = B/(t-2)B \subset K^d, \qquad O = \mathcal{O}_K^d. \tag{2}$$

The inclusion in K^d uses the d distinct points of the split fibre. Every element of C is integral over $\mathcal{O}_K$, so $C \subset O$. Let $\mathfrak{D}(C)$ be the discriminant ideal of its trace pairing.

Theorem 1.1. *With this notation,*

$$h_{\mathrm{F}}(\operatorname{Jac} X) \leq \frac{\log \mathrm{N}\mathfrak{D}(C)}{2[K:\mathbb{Q}]} + g \log \frac{2}{\sqrt{\pi}}. \tag{3}$$

The auxiliary field K is used to split a finite fibre and to compute a stable height. Its degree will cancel from the estimates. No upper bound for $[K:\mathbb{Q}]$ is assumed in this theorem.

2 Trace-dual lattices and regular differentials

Lemma 2.1. *Let $C \subset O = \mathcal{O}_K^d$ be an order with $C \otimes K = K^d$, and let*

$$C^\vee = \{x \in K^d : \operatorname{Tr}_{K^d/K}(xC) \subset \mathcal{O}_K\}.$$

For any set I of g coordinate positions, set $F = \operatorname{pr}_I(C^\vee) \subset K^g$. Then F is a projective $\mathcal{O}_K$-module containing $\mathcal{O}_K^g$. With the standard Euclidean metric on K^g at every embedding,

$$\widehat{\deg}(F) \leq \tfrac{1}{2} \log \mathrm{N}\mathfrak{D}(C).$$

Proof. Work locally at a finite prime $\mathfrak{p}$. The trace pairing on K^d is the ordinary dot product, and O is self-dual. Thus $O \subset C^\vee$, and dualizing the inclusion $C \subset O$ shows

$$\operatorname{length}(C^\vee/O) = \operatorname{length}(O/C) = \tfrac{1}{2} v_{\mathfrak{p}}(\mathfrak{D}(C)).$$

For example, if a basis matrix for C relative to O is U, then the trace Gram matrix is $U^{\mathsf{T}}U$, which proves the last equality. These equalities are valid locally even when C is not globally free.

Projection gives a surjection $C^\vee/O \twoheadrightarrow F/\mathcal{O}_K^g$. The latter quotient therefore has no larger length at any prime. A finite torsion-free module over the Dedekind ring $\mathcal{O}_K$ is projective. Finally, the standard basis of $\mathcal{O}_K^g$ has determinant norm one, so

$$\widehat{\deg}(F) = \log \#(F/\mathcal{O}_K^g) \leq \tfrac{1}{2} \log \mathrm{N}\mathfrak{D}(C).$$

□

Lemma 2.2. *Let $\mathcal{X}$ be a minimal regular semistable model of X. The evaluations of a differential in $E = H^0(\mathcal{X}, \omega_{\mathcal{X}/\mathcal{O}_K})$ at the points of $\pi^{-1}(2)$, expressed in the coordinate t, lie in $C^\vee$. Equivalently, there is an integral map*

$$E \longrightarrow C^\vee, \qquad \omega \longmapsto \left(\frac{\omega}{dt}(P)\right)_{P \in \pi^{-1}(2)}.$$

Proof. First, normalization is finite because the base is excellent. A normal surface is Cohen–Macaulay, and a finite morphism from such a surface to a regular surface is flat by miracle flatness; see [1, Tag 00R4]. Consequently B is finite projective over $\mathcal{O}_K[t]$.

Take a common regular model $\mathcal{Z}$ dominating $\mathcal{X}$ and $\mathcal{Y}$. Pullback under a blowup of regular arithmetic surfaces takes regular relative differentials to regular relative differentials. One can also read this directly from the canonical-sheaf formula for a blowup; see [2, Lemma 2.3.1]. Hence a differential in E is regular on $\mathcal{Z}$. The birational morphism $\mathcal{Z} \to \mathcal{Y}$ is an isomorphism at every codimension-one point of the normal surface $\mathcal{Y}$.

The dualizing module of a Cohen–Macaulay normal surface is reflexive: it is torsion-free and satisfies Serre's condition (S_2). Thus sections defined at all codimension-one points extend across the codimension-two complement; see [1, Tags 0AWQ and 0AVT, Lemmas 31.13.12–31.13.14]. The differential therefore defines a section of the relative dualizing sheaf of $\mathcal{Y}$.

Finite duality identifies this sheaf on the affine chart with

$$\operatorname{Hom}_{\mathcal{O}_K[t]}(B, \mathcal{O}_K[t])\, dt;$$

see [1, Tags 0BUL and 0C0I]. This statement uses the dualizing *module*; it does not require that the normal surface have an invertible dualizing sheaf. Because B is finite projective, specializing the dual at $t = 2$ gives $\operatorname{Hom}_{\mathcal{O}_K}(C, \mathcal{O}_K)\, dt$. On the generic fibre the identification sends a differential $f\, dt$ to the functional $b \mapsto \operatorname{Tr}(fb)$. After splitting the fibre, its coordinates are exactly the values $f(P) = \omega/dt(P)$. This proves the assertion. □

Proof of Theorem 1.1. Riemann–Hurwitz gives

$$2g - 2 = -2d + \deg \operatorname{Ram}(\pi) \le d - 3,$$

since each of the three branch fibres contributes at most $d - 1$. In particular $d > 2g - 2$. A nonzero holomorphic differential cannot vanish at all d distinct points of $\pi^{-1}(2)$. Therefore the evaluation map has rank g. Choose g coordinate positions I for which projection remains injective. By Lemmas 2.1 and 2.2, we obtain an integral, generically invertible map

$$\phi : E \longrightarrow F = \operatorname{pr}_I(C^\vee), \qquad \widehat{\deg}(F) \le \tfrac{1}{2} \log \mathrm{N}\mathfrak{D}(C).$$

Fix a complex embedding of K. The disc $D = \{t : |t - 2| < 1/2\}$ is simply connected and avoids the branch locus. Its inverse image is a disjoint union of d analytic sheets, each mapped isomorphically to D. On one such sheet write $\omega = f(t)\, dt$. The mean-value inequality implies

$$|f(2)|^2 \le \frac{4}{\pi} \int_D |f(t)|^2\, dx\, dy \le \frac{4}{\pi} \|\omega\|_{L^2}^2.$$

Hence the norm of each evaluation functional is at most $2/\sqrt{\pi}$. In an orthonormal basis of $E \otimes_\sigma \mathbb{C}$, each row of the evaluation matrix has this same bound on its Euclidean norm. Hadamard's determinant inequality gives

$$\| \det \phi \|_\sigma \le (2/\sqrt{\pi})^g.$$

The determinant is nonzero and integral. Taking its arithmetic degree yields

$$\widehat{\deg}(E) \le \widehat{\deg}(F) + \sum_\sigma \log \| \det \phi \|_\sigma \le \tfrac{1}{2} \log \mathrm{N}\mathfrak{D}(C) + [K : \mathbb{Q}] g \log(2/\sqrt{\pi}).$$

Now apply (1). □

3 The discriminant of the fibre

Let S be a finite set of rational primes, and put

$$R_S = \prod_{p \in S} p, \qquad \omega(S) = \#S.$$

Assume that the normalization defining the Belyi cover is finite étale over

$$\operatorname{Spec} \mathcal{O}_{K_0}[S^{-1}, t, t^{-1}, (t-1)^{-1}], \tag{4}$$

where K_0 is a field of definition. This property persists after any finite field extension: finite étale base change is finite étale, and a finite étale algebra over a normal base is normal. Thus it agrees with the normalization of the new generic cover on this open set. In particular the auxiliary splitting field in Section 1 introduces no new vertical branch primes outside S.

Lemma 3.1. *Under (4), the order C in (2) satisfies*

$$\frac{\log \mathrm{N}\mathfrak{D}(C)}{[K:\mathbb{Q}]} \le d\left(\log 2 + \log R_S + \omega(S)\log d\right).$$

Proof. The determinant of the trace pairing of $B/\mathcal{O}_K[t]$ defines an effective discriminant divisor on the base. Formation of the trace pairing commutes with specialization. Its intersection with the section $t = 2$ is therefore $\log \mathrm{N}\mathfrak{D}(C)$. The coefficient of this divisor at a codimension-one point is $\sum_j f_j r_j$, where e_j, f_j, r_j are the ramification index, residue degree, and different exponent of the corresponding extensions of discrete valuation rings. They satisfy $\sum_j e_j f_j = d$. These are the usual local trace-discriminant and different identities; see [3, Chapter III, Sections 3 and 6] and [2, Section 4.2].

At a horizontal branch divisor the residue characteristic is zero. The different exponent is $r_j = e_j - 1$, so the discriminant coefficient is at most d. The section $t = 2$ meets $t = 0$ with normalized intersection $\log 2$, and meets neither $t = 1$ nor $t = \infty$. Thus the horizontal contribution is at most $d \log 2$.

Let $\mathfrak{p} \mid p$ be a prime of K, and write $e_\mathfrak{p} = v_\mathfrak{p}(p)$. Lenstra's different bound, in the form [2, Proposition 4.1.3], gives

$$r_j \le e_j - 1 + e_j e_\mathfrak{p} \left\lfloor \frac{\log d}{\log p} \right\rfloor.$$

This bound also applies to the possibly imperfect residue field at the generic point of a vertical fibre. Its discriminant coefficient is consequently at most

$$d\left(1 + e_\mathfrak{p} \left\lfloor \frac{\log d}{\log p} \right\rfloor\right).$$

Only $p \in S$ can contribute. The intersection of the vertical fibre over $\mathfrak{p}$ with the section has value $f_\mathfrak{p} \log p$. Using

$$\sum_{\mathfrak{p} \mid p} e_\mathfrak{p} f_\mathfrak{p} = [K:\mathbb{Q}], \qquad \sum_{\mathfrak{p} \mid p} f_\mathfrak{p} \le [K:\mathbb{Q}],$$

and $\lfloor \log d / \log p \rfloor \log p \le \log d$ proves the estimate. □

Theorem 3.2. *For a Belyi cover satisfying (4),*

$$h_\mathrm{F}(\operatorname{Jac} X) \le \frac{d}{2}\left(\log 2 + \log R_S + \omega(S)\log d\right) + g \log \frac{2}{\sqrt{\pi}}. \tag{5}$$

Proof. Combine Theorem 1.1 and Lemma 3.1. □

4 The integral modular cover

Let N be a positive integer and let $H \subset (\mathbb{Z}/N\mathbb{Z})^\times$ contain -1. Write

$$X_H = X_1(N)/H, \qquad t_H = [(\mathbb{Z}/N\mathbb{Z})^\times : H], \qquad J_H = \operatorname{Jac} X_H.$$

The action is the diamond action. Assume that X_H has positive genus, and use the rational map $t = j/1728$ to the projective line.

Lemma 4.1. *The map $X_H \to \mathbb{P}^1_t$ has degree*

$$d_H = N t_H \prod_{p \mid N} (1 + p^{-1}).$$

It is unramified outside $0, 1, \infty$ and admits a finite étale model over

$$U = \operatorname{Spec} \mathbb{Z}[1/(6N), t, t^{-1}, (t-1)^{-1}].$$

Proof. Over U consider

$$E_t : \quad y^2 = x^3 - \frac{3t}{t-1} x - \frac{2t}{t-1}. \tag{6}$$

Its discriminant and j-invariant are

$$\Delta(E_t) = \frac{1728 t^2}{(t-1)^3}, \qquad j(E_t) = 1728 t.$$

Thus it is an elliptic curve over U. Since N is invertible, its scheme of points of exact order N is finite étale. The constant group H acts freely on this scheme: if $aP = P$ and P has exact order N, then $a = 1$ in $(\mathbb{Z}/N\mathbb{Z})^\times$. The quotient is finite étale. This follows also from the classification of finite étale covers by locally constant finite sets; freeness is enough, even when a residue characteristic divides $|H|$.

To identify its generic fibre, fix an elliptic curve with j-invariant different from 0 and 1728 over an algebraic closure. Its automorphism group is $\{1, -1\}$. The geometric fibre of $X_1(N)$ is the set of primitive torsion points modulo this automorphism group. Since $-1 \in H$, the fibre of X_H is precisely the set of primitive torsion points modulo H. Different isomorphisms to (6) differ by ± 1 and hence give the same H-orbit. In particular quadratic twisting introduces no extra double cover. This identifies the finite étale model with the normalization of X_H over U.

There are $N^2 \prod_{p \mid N} (1 - p^{-2})$ primitive points. Division by $|H| = \varphi(N)/t_H$ gives the asserted degree. Equivalently, the congruence subgroup contains $-I$, so its $\mathrm{SL}_2(\mathbb{Z})$ index is already the degree of the coarse map to $X(1)$; see [4, Section 4]. The only possible branch points of that map are the two elliptic points and the cusp, which become $0, 1, \infty$ in the t coordinate. □

Lemma 4.2. *Let V be an excellent discrete valuation ring of characteristic zero, let E/V be an elliptic scheme, and let n be a positive integer. Write $E[n] = \operatorname{Spec} A$, and let $\widetilde{A}$ be its normalization in its generic finite étale algebra. Under the generic trace pairing, the complementary modules satisfy*

$$A^\vee = n^{-1} A, \qquad \widetilde{A}^\vee \subseteq n^{-1} \widetilde{A}.$$

The different of every field component of $\widetilde{A}/V$ therefore contains n. The same assertion holds for every intermediate field of such a component.

Proof. Multiplication $[n] : E \to E$ is finite flat and a local complete intersection morphism. Its derivative in invariant differential bases is multiplication by n. Consequently its relative differential module has zeroth Fitting ideal (n). For a finite flat local complete intersection, this Kähler different is the different; see [1, Tag 0BWB, Lemmas 49.12.3 and 49.12.6].

Base change along the zero section gives $E[n]/V$. The relative differential module and its zeroth Fitting ideal commute with base change, while the morphism remains finite flat and a local complete intersection. Its different is therefore (n). Its relative dualizing module is invertible. Under the trace pairing this is the complementary fractional ideal $A^\vee$, whose inverse is the different; see [1, Tag 0BW0, Lemmas 49.8.2–49.8.3, and Tag 0BWB, Lemma 49.12.4]. Thus $A^\vee = n^{-1}A$ as fractional ideals, not merely as abstract modules.

Normalization is finite, and inclusion reverses trace duals:

$$\widetilde{A}^\vee \subseteq A^\vee = n^{-1}A \subseteq n^{-1}\widetilde{A}.$$

Hence $n\widetilde{A}^\vee \subseteq \widetilde{A}$, which says exactly that the different contains n. Finally, for normal discrete valuation rings $V \subseteq W \subseteq U$, the different formula

$$\mathfrak{D}_{U/V} = \mathfrak{D}_{U/W}\mathfrak{D}_{W/V}U$$

shows that $v_U(\mathfrak{D}_{U/V}) \leq v_U(n)$ implies $v_W(\mathfrak{D}_{W/V}) \leq v_W(n)$, because the different exponent of U/W is nonnegative. Apply this in each field component and at each place. These finite-duality and different arguments do not require a perfect residue field. □

Corollary 4.3. *Let g_H be the genus of X_H. Then*

$$h_{\mathrm{F}}(J_H) \leq \frac{d_H}{2}\left(\log 12 + \log N + 2\log d_H\right) + g_H \log\frac{2}{\sqrt{\pi}}. \tag{7}$$

Proof. Use the auxiliary field K from Theorem 1.1. At a prime $\mathfrak{p}$ of K over $p > 3$, localize the t-line at the generic point of the special fibre. The family (6) is an elliptic scheme over this discrete valuation ring: its discriminant $1728t^2/(t-1)^3$ is a unit there. By Lemma 4.1, the generic cover for X_H is the primitive N-torsion cover modulo H. Each of its field components is an intermediate field of a field generated by a single N-torsion point. Lemma 4.2 therefore bounds every different exponent above $\mathfrak{p}$ by the valuation of N. The coefficient of the vertical discriminant divisor is at most

$$d_H v_{\mathfrak{p}}(N).$$

After summing over these primes and dividing by $[K:\mathbb{Q}]$, the contribution is at most $d_H \log N$.

At $p = 2, 3$, the estimate in Lemma 3.1 gives at most $d_H(\log 6 + 2\log d_H)$. The horizontal intersection at $t = 2$ contributes at most $d_H \log 2$. Thus the normalized logarithmic discriminant of the fibre order is at most

$$d_H(\log 12 + \log N + 2\log d_H).$$

Apply Theorem 1.1. In particular no degree of a full torsion splitting field enters this estimate. □

5 An abelian surface occurring as an isogeny factor

Lemma 5.1. *Let J be an abelian variety of dimension $g \geq 2$ over a number field L', and let A be an abelian surface which is an isogeny factor of J over L'. Put $d' = [L':\mathbb{Q}]$ and*

$$H_J = \max(1, h_{\mathrm{F}}(J) + 3g/2).$$

Then

$$h_{\mathrm{F}}(A) \leq H_J + 8\left(32\log 14 + \log d' + \log\max(1, \log d', H_J)\right). \tag{8}$$

Proof. Choose an L'-homomorphism $A \to J$ with finite kernel, and let $A_0 \subset J$ be its image. It is an abelian surface over L' and is L'-isogenous to A. The stable version of Rémond's subvariety inequality [5, Corollary 1.2 and the paragraph following Corollary 1.3] gives

$$h_{\mathrm{F}}(A_0) + h_{\mathrm{F}}(J/A_0) \le h_{\mathrm{F}}(J).$$

Bost's lower bound in the original normalization is $h_{\mathrm{F}}(B) \ge -\log(\pi\sqrt{2})\dim B > -3\dim B/2$. Thus $h_{\mathrm{F}}(A_0) \le H_J$.

Apply [6, Theorem 1.9(1)] with source A_0 and target A, both over L'. In dimension two it supplies an isogeny of degree at most

$$\left(14^{32} d' \max(1, \log d', h_{\mathrm{F}}(A_0))\right)^{16}.$$

Stable Faltings height increases under an isogeny of degree D by at most $\frac{1}{2}\log D$; see also [5, Proposition 2.1]. Replacing $h_{\mathrm{F}}(A_0)$ by H_J proves (8). □

The field L' in this lemma is the field on which the surface factor is realized. It need not split a fibre of a modular curve, and it need not be the auxiliary field K of Theorem 1.1. The evaluation estimate is a stable height bound valid after arbitrary extension; the degree entering (8) is only $[L' : \mathbb{Q}]$.

6 The five-contact height estimate

For $a = u/v \in \mathbb{Q}$ in lowest terms, with $v > 0$, put $h(a) = \log\max(|u|, v)$. Define

$$S_* = \{p \text{ prime} : p \le 76176\}, \quad B_* = \prod_{p \in S_*} p, \quad \mathcal{D} = \{0, 1, 2401, -343/32\}.$$

For $a \notin \mathcal{D}$, let $M_5(a)$ be the product of the primes $p \notin S_*$ for which at least one of

$$v_p(v), \quad v_p(u), \quad v_p(u - v), \quad v_p(u - 2401v), \quad v_p(32u + 343v)$$

is positive and odd. Every prime occurs only once. The first valuation is the contact at infinity.

We use the following arithmetic input from [7, proof of `o8-height`, equations (LR2)–(LR5), and proof of `quartic-height19`].

Definition 6.1 (Modular realization input)**.** For each $a \in \mathbb{Q} \setminus \mathcal{D}$, put $R = B_* M_5(a)$. There is an abelian surface A such that

$$h(a) \le 2\max(0, h_{\mathrm{F}}(A)) + 128. \tag{9}$$

Either $h_{\mathrm{F}}(A) < 854R$, or there are a normal number field $L'/\mathbb{Q}$ and an even nebentypus ε of a primitive weight-two newform, of level N, such that

$$\begin{aligned} [L' : \mathbb{Q}] \le 2304\varphi(R), \qquad N \le 210^{300} R^2, \qquad \operatorname{rad}(N) \mid R, \\ \varepsilon^2|_{G_{L'}} = 1, \qquad A_{L'} \text{ is an isogeny factor of } \operatorname{Jac}(X_1(N)/\ker\varepsilon)_{L'}. \end{aligned} \tag{10}$$

Here R is squarefree and divisible by 6. The degree bound in (10) retains the totient from the scalar character: its order satisfies $m \le 2\prod_{p \mid R,\ p \text{ odd}}(p-1) = 2\varphi(R)$, and the construction gives $[L' : \mathbb{Q}] \le 1152m$. The square of the nebentypus is trivial over L' by the determinant identity for the compatible representations. The newform quotient factors through the diamond quotient because every element of $\ker\varepsilon$ fixes all conjugates of its differentials. No bound on the degree of the newform's coefficient field is assumed.

Theorem 6.2. *Assume the modular realization input. Put*

$$D_0 = 4608 \cdot 210^{300}, \qquad C_3 = 19 D_0 B_*^3 \log B_*.$$

Then $C_3 < 2^{330770}$ *and*

$$h(a) \le C_3 M_5(a)^3 (1 + \log M_5(a)) \quad (a \in \mathbb{Q} \setminus \mathcal{D}). \tag{11}$$

In particular,

$$h(a) < 2^{330789} M_5(a)^{7/2}. \tag{12}$$

For every fixed $\alpha > 3$ *there is an effective constant* C_α *such that* $h(a) \le C_\alpha M_5(a)^\alpha$.

Proof. Set $M = M_5(a)$ and $R = B_* M$. In the non-CM alternative, let $H = \ker \varepsilon$ and $t_H = \operatorname{ord} \varepsilon$. Since ε^2 is trivial on $G_{L'}$, its image order divides $[L' : \mathbb{Q}]$. Hence

$$t_H \le 2[L' : \mathbb{Q}] \le 4608 \varphi(R).$$

The exact index identity in Lemma 4.1 gives

$$\begin{aligned} d_H &\le 4608 \cdot 210^{300} R^2 \varphi(R) \prod_{p \mid R} (1 + p^{-1}) \\ &= D_0 R^3 \prod_{p \mid R} (1 - p^{-2}) < D_0 R^3. \end{aligned} \tag{13}$$

Also $2 \le g_H \le d_H$, because the Jacobian contains a surface factor and the cover is a Belyi map.

Write $\ell = \log R$. The exact prime-product computation in Section 8 gives

$$D_0^2 < B_*, \qquad 144 < B_*, \qquad \ell \ge 1.$$

Consequently

$$\log N \le \tfrac{5}{2}\ell, \qquad \log d_H \le \tfrac{7}{2}\ell, \qquad \log 12 \le \tfrac{1}{2}\ell.$$

Using $\log(2/\sqrt{\pi}) < 1$ in Corollary 4.3, we obtain

$$h_{\mathrm{F}}(J_H) \le 5 d_H \ell + d_H \le 6 d_H \ell. \tag{14}$$

Thus the number H_J in Lemma 5.1 is at most

$$H_* = 8 D_0 R^3 \ell.$$

Let $d' = [L' : \mathbb{Q}]$. Since $d' \le 2304 R$, $2304 < e^8$, $\ell \ge 1$, and $\log D_0 \le \ell$,

$$\log d' < 9\ell, \qquad \log H_* \le 9\ell.$$

Moreover $H_* > \max(1, \log d')$. Lemma 5.1, with $\log 14 < 3$, now gives

$$\begin{aligned} h_{\mathrm{F}}(A) &\le H_* + 8(96 + 18\ell) \\ &\le H_* + 912\ell \le 9 D_0 R^3 \ell. \end{aligned}$$

The CM bound $854R$ is smaller than the same expression. Combining with (9), and using $D_0 R^3 \ell > 128$, gives

$$h(a) \le 19 D_0 R^3 \ell.$$

Finally $\log R = \log B_* + \log M \le (\log B_*)(1 + \log M)$, since $\log B_* > 1$ and $M \ge 1$. This proves (11).

The explicit numerical bound uses

$$19 < 2^5, \qquad D_0 < 2^{2413}, \qquad B_* < 2^{109445}, \qquad \log B_* < 2^{17}.$$

Consequently

$$C_3 < 2^{5 + 2413 + 3 \cdot 109445 + 17} = 2^{330770}.$$

For $u \ge 0$, the maximum of $(1+u)e^{-u/2}$ occurs at $u = 1$ and is less than two. Taking $u = \log M$ gives a constant smaller than 2^{330771}, and in particular proves the convenient bound (12). More generally, for each $\eta > 0$, $(1+u)e^{-\eta u}$ is bounded on $[0, \infty)$ by an explicitly computable number. Apply this with $\eta = \alpha - 3$. □

7 Disjoint sets of contacts

Corollary 7.1. *Fix $r \geq 1$ and put*

$$\Theta_r = \bigcup_{j=0}^{r-1} \left(3j + \left\{0, -\tfrac{1}{2}, -1, \tfrac{1}{2399}, -\tfrac{32}{407}\right\}\right).$$

There is an explicitly determined finite exceptional set Σ_r such that, if $M_r(t)$ is the product of primes outside Σ_r at which some $t - \theta$, $\theta \in \Theta_r$, has positive odd valuation, then

$$h(t) \ll_r M_r(t)^{3/r}(1 + \log M_r(t)) \ll_r M_r(t)^{7/(2r)} \qquad (t \in \mathbb{Q} \setminus \Theta_r).$$

All implied constants are effective under the stated modular realization input.

Proof. Apply Theorem 6.2 to $\phi_j(t) = 2 + 1/(t - 3j)$ for $0 \leq j < r$. Their matrices have determinant -1, so $h(t) \leq h(\phi_j(t)) + O_r(1)$. In addition to S_*, exclude the primes dividing contact coefficients and the nonzero determinants of pairs of primitive contact forms. The $5r$ rational contacts are distinct. Outside the resulting fixed set, different contact forms cannot vanish at the same residue point. Thus the radicals $M_j(t)$ for the different blocks have disjoint prime support, and $\prod_j M_j(t) = M_r(t)$. Removing the exceptional primes from a block changes its height inequality by at most an effective fixed factor.

Multiplying the r inequalities gives

$$h(t)^r \ll_r M_r(t)^3 \prod_{j=0}^{r-1}(1 + \log M_j(t)).$$

Each logarithmic factor is at most $1 + \log M_r(t)$. Taking rth roots proves the first bound. Applying (12) in each block proves the second. The primitive contact forms, their pairwise determinants, and the fixed exceptions can be constructed directly as in [8, Sections 1–2]. □

Four translated sets of five contacts therefore give height exponent $7/8$ with twenty distinct finite contacts. This statement concerns logarithmic height; replacing it by an estimate for multiplicative height would be a different assertion.

8 Exact verification

The script

`computation/contact_heights/check_evaluation_height.py`

verifies the rational formulas for the discriminant and j-invariant in (6), reconstructs B_* by a sieve with independent trial division, and checks the integer comparisons $D_0^2 < B_*$, $144 < B_*$, $D_0 < 2^{2413}$, and $B_* < 2^{109445}$. It also checks the exponent arithmetic and the exponential inequalities used above by rational partial sums. Its output is `evaluation-height-certificate.json` in the same directory. The finite-algebra and geometric assertions are proved in the text; they are not inferred from numerical experiments.

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